Clarity Without Convergence: Commercial SATCOM and the Information Contest over Participation in the Russia–Ukraine War

Santiago Verace - z5223432

s.verace@student.unsw.edu.au

ADFA / University of New South Wales – ZEIT8115 S2_2026

15 October 2026

Word count (body, excluding abstract, footnotes and bibliography): 3852

Abstract

Commercial SATCOM created two different problems in Ukraine, and they were often discussed as though they were one. KA-SAT raised a technical and legal problem about what part of a commercial system was actually being used, affected or potentially targetable. Starlink raised another problem because Ukrainian forces could depend on a service whose coverage and permitted uses remained under private control. Russia and SpaceX were not answering quite the same question. Russia's October 2022 statement treated Western military reliance on commercial space infrastructure as "indirect participation in military conflicts". SpaceX later spoke from the provider side: military communications could continue, but offensive use fell outside the line it had drawn. Making both positions public did not turn them into a common test.

Over time, more of that disagreement entered the public record. Article 52 still required an object-specific assessment, regardless of the broader labels being used around commercial participation. Wiener helps explain why participation could take on content through practice while its meaning remained contested. Fearon leaves a different gap. Actors can state a position publicly while keeping important information, incentives and calculations private. The case therefore does not show transparency producing convergence. It shows a disagreement becoming easier to identify.

Introduction

Commercial SATCOM was already part of the war in its first weeks. On 24 February 2022, a cyber operation disrupted Viasat's KA-SAT network. The United Kingdom later attributed the operation to Russia and assessed that the Ukrainian military was believed to have been the primary target.[1] The United States described the same event in more operational terms, saying the attack was intended to disrupt Ukrainian command and control.[2] By October, however, the argument had moved beyond who carried out the attack. Vorontsov used the UN First Committee to describe Western military use of civilian and commercial space infrastructure as "indirect participation in military conflicts" and warned that "Quasi-civilian infrastructure may become a legitimate target for retaliation."[3] SpaceX put a different boundary into the public record in February 2023. Shotwell drew the line differently. Using Starlink for military communications was acceptable, she said, but offensive use was not; SpaceX had already taken steps to restrict it.[4]

Authority over the service never sat neatly with the actor using it. A military could depend on the network without owning it, and SpaceX could still decide where coverage existed or which uses fell outside its agreement. States, meanwhile, could attach consequences to the same relationship that neither provider nor user accepted. The question examined here is: How did competing public articulations by states and commercial providers shape the information contest over commercial SATCOM in the Russia–Ukraine war, and what does this reveal about the capacity of transparency to reduce interpretive ambiguity?

Starlink brings the dispute into the provider relationship itself. Ukrainian forces were using the service, yet SpaceX still decided which forms of military use it would permit. Russia could look at the same military reliance on commercial infrastructure

---

1 UK National Cyber Security Centre, "Russia behind Cyber Attack with Europe-Wide Impact an Hour before Ukraine Invasion," 10 May 2022, https://www.ncsc.gov.uk/news/russia-behind-cyber-attack-with-europe-wide-impact-hour-before-ukraine-invasion.
2 Antony J. Blinken, "Attribution of Russia's Malicious Cyber Activity Against Ukraine," U.S. Department of State, 10 May 2022, https://2021-2025.state.gov/attribution-of-russias-malicious-cyber-activity-against-ukraine/.
3 Konstantin Vorontsov, "Statement by Deputy Head of the Russian Delegation Mr. Konstantin Vorontsov at the Thematic Discussion on Outer Space (Disarmament Aspects) in the First Committee of the 77th Session of the UNGA," Permanent Mission of the Russian Federation to the United Nations, 26 October 2022, unofficial translation, https://russiaun.ru/en/news/261022_v.
4 Jeff Foust, "Shotwell: Ukraine 'Weaponized' Starlink in War against Russia," *SpaceNews*, 8 February 2023, https://spacenews.com/shotwell-ukraine-weaponized-starlink-in-war-against-russia/.

and describe it as participation. Both positions could be made public without answering the question in the same way.

Kuplic and Sawmiller approach one part of this problem through targeting law and Viasat. Their distinction between present military "use" and future "purpose" shows one way a commercial object can enter military-objective analysis.[5] They also place weight on what states make visible. Published policy, practice and dialogue can show how existing obligations are being read without requiring every operational threshold to be disclosed. The difficulty is that visibility is not the same thing as acceptance. West and Azcárate Ortega argue that norms still depend on implementation and trust; where those conditions are weak, publication may simply give competing interpretations a clearer form.[6]

Starlink sat in that gap. The competing positions were public, but the point at which commercial support became participation remained contested.

Commercial SATCOM in the Information Environment

JP 3-04 changed the vocabulary without making the commercial network disappear from the problem. It replaced JP 3-13 and dropped the older language of Information Operations and information-related capabilities.[7] What matters here is where the network now sits. Current US doctrine places systems, organisations and people together inside the information environment, so a commercial communications service becomes part of the setting in which information is collected, moved, interpreted and acted upon. The 2023 Department of Defense Strategy for Operations in the Information Environment defines the information environment as "the aggregate of social, cultural, linguistic, psychological, technical, and physical factors that affect how humans and automated systems derive meaning from, act upon, and are impacted by information, including the individuals, organizations, and

5 G. Blair Kuplic and Jonathan Sawmiller, "Humanity on the Final Frontier: Challenges in Applying International Humanitarian Law to Modern Military Space Operations," *International Review of the Red Cross* 107, no. 928 (2025): 219, 235–37, https://doi.org/10.1017/S1816383124000559.
6 Jessica West and Almudena Azcárate Ortega, *Norms for Outer Space: A Small Step or a Giant Leap for Policymaking?*, Space Dossier 7 (Geneva: United Nations Institute for Disarmament Research, 2022), 26–27, https://doi.org/10.37559/WMD/22/Space/01.
7 Congressional Research Service, *Defense Primer: Operations in the Information Environment*, IF10771, updated 29 November 2024, https://www.everycrsreport.com/reports/IF10771.html.

systems that collect, process, disseminate, or use information."[8] Commercial SATCOM fits comfortably inside it: satellites and terminals matter, but so do gateways, management networks, operators and access rules. The same Strategy describes consistent integration of informational and physical power as the norm rather than treating information as a separate messaging function.

JP 3-13 had offered a cleaner division of the environment. Communications networks, platforms and key decision-makers occupied the physical dimension; collection, processing, storage and dissemination the informational; perception, judgement and decision-making the cognitive.[9] JCOIE criticised the older model for putting too much weight on transmission. The criticism was about interpretation as much as transmission. Audiences bring different worldviews and contexts to the same information, so receipt of a message does not tell us what meaning will be attached to it.[10] Starlink adds a control problem. Ukrainian forces could use the service while SpaceX retained decisions over coverage and acceptable use, and civilians remained on the same infrastructure. The network could therefore function normally while authority over the service was divided.

By October 2022, military use was already visible. Russia's statement mattered because it attached a consequence to that use rather than revealing it. Wither is useful at that point: ambiguity can shape a response even when the activity itself is known. Disinformation and destabilisation can leave leaders unsure about what they are looking at or what they should do next.[11]

Later evidence is kept separate from what actors could know at the time. A disclosure made months afterwards can change what can be reconstructed about an event. It does not change when the information became public. Rid and Buchanan treat attribution as assessment and communication together, including communication to third parties and the public.[12] Paul cautions against treating later

8 U.S. Department of Defense, *Strategy for Operations in the Information Environment* (Washington, DC: Department of Defense, 2023), 1.
9 Joint Chiefs of Staff, *Joint Publication 3-13: Information Operations* (Washington, DC: Joint Chiefs of Staff, 27 November 2012, incorporating Change 1, 20 November 2014), I-2–I-3.
10 Joint Chiefs of Staff, Joint Concept for Operating in the Information Environment (JCOIE) (Washington, DC: Joint Chiefs of Staff, 25 July 2018), 3–4.
11 James K. Wither, "Making Sense of Hybrid Warfare," *Connections: The Quarterly Journal* 15, no. 2 (2016): 82, https://doi.org/10.11610/Connections.15.2.06.
12 Thomas Rid and Ben Buchanan, "Attributing Cyber Attacks," *Journal of Strategic Studies* 38, nos. 1–2 (2015): 7, https://doi.org/10.1080/01402390.2014.977382.

behaviour as evidence that an earlier message worked when other causes remain plausible.[13]

"Public articulation" and "transparency" are not used as synonyms here. "Public articulation" means an open statement of a position, interpretation or warning; "transparency" is the stronger proposition that making policies and interpretations public can reduce misunderstanding or miscalculation. Deppe and Schaal's warning about conceptual stretching is useful here because a category broad enough to absorb every informational consequence stops distinguishing very much.[14] The evidence is public and mostly English-language, which leaves classified targeting assessments and SpaceX's internal deliberations outside it. The Russian Mission statement is assessed through its published unofficial English translation.

Law, Interpretation and the Norms Proposition

Russia's language of participation and retaliation was politically broad, whereas targeting law asks a narrower set of questions. Article 52(2) of Additional Protocol I requires both an effective contribution to military action by nature, location, purpose or use and a definite military advantage from destruction, capture or neutralisation in the circumstances ruling at the time.[15] The requirements are cumulative. Article 52(3) adds a presumption where an object normally dedicated to civilian purposes is in doubt, while Article 57 keeps verification, feasible precautions and proportionality in the analysis even after an object is considered a military objective.[16]

Article 52 does not create a third category between civilian object and military objective. An object remains protected as civilian unless the military-objective test is satisfied in the circumstances ruling at the time. Berrang reaches the issue from another direction. The fact that a commercial system serves both civilian and military users does not, in his view, make a new space-specific Geneva Convention

13 Christopher Paul, *Strategic Communication: Origins, Concepts, and Current Debates* (Santa Barbara, CA: Praeger, 2011), 119.
14 Christoph Deppe and Gary S. Schaal, "Cognitive Warfare: A Conceptual Analysis of the NATO ACT Cognitive Warfare Exploratory Concept," *Frontiers in Big Data* 7 (2024): 2, 10, https://doi.org/10.3389/fdata.2024.1452129.
15 Protocol Additional to the Geneva Conventions of 12 August 1949, and Relating to the Protection of Victims of International Armed Conflicts (Protocol I), 8 June 1977, 1125 U.N.T.S. 3, art. 52(2)–(3).
16 Additional Protocol I, arts. 52(3), 57.

necessary.[17] Hathaway, Khan and Revkin focus on a different risk. If "dual-use" begins functioning as a legal category of its own, the civilian–military distinction can blur before the object-specific test is actually applied. Their concern is that civilian protection can weaken when the label starts doing work that Article 52 assigns to the legal test.[18]

Calling a satellite network one object hides most of the targeting problem. A satellite, a gateway, a terminal and the management segment do not necessarily perform the same function, and civilian and military dependence can sit on different parts of the same service. Kuplic and Sawmiller's distinction between present "use" and future "purpose" matters at that level. Present military use can be identified from what an object is doing, whereas a claim about future military purpose ordinarily requires more specific intelligence; their example is Ukrainian reliance on secure Viasat communications.[19]

Military communications may make an effective contribution to military action, but that does not dispense with the separate requirement for definite military advantage. The Woomera Manual also keeps distinguishable civilian components and continuing verification in view, making the problem harder to collapse into the label "satellite network." [20]

Bi and Zhu work at a smaller scale. Their analysis begins with the smallest feasible function of military relevance and then asks what civilian dependence and cascading effects sit around it.[21] In some systems, that leaves the relevant question at the level of a function or component rather than the network as a whole.

Publishing those facts and policies can still matter because other actors then have more to work with than inference alone. Kuplic and Sawmiller link guidance, state practice and dialogue to the possibility of common understandings and fewer

---

17 Svenja Berrang, "Does the Dual-Use of Space Objects Necessitate a New Geneva Convention?," *Case Western Reserve Journal of International Law* 57 (2025): 315, 341, https://scholarlycommons.law.case.edu/jil/vol57/iss1/22/.

18 Oona A. Hathaway, Azmat Khan, and Mara R. Revkin, "The Dangerous Rise of 'Dual-Use' Objects in War," *Yale Law Journal* 134, no. 8 (2025): 2652–53, https://yalelawjournal.org/article/the-dangerous-rise-of-dual-use-objects-in-war.

19 Kuplic and Sawmiller, "Humanity on the Final Frontier," 219.

20 Jack Beard and Dale Stephens, eds., with David Koplow, *The Woomera Manual on the International Law of Military Space Activities and Operations* (Oxford: Oxford University Press, 2024), 320–21, 331–32, https://doi.org/10.1093/law/9780192870667.001.0001.

21 Chen Bi and Yiyang Zhu, "From Objects to Interdependence: How the Law of Armed Conflict Regulates Attacks on Dual-Use Information Infrastructure in Cyber and Information Warfare," *Cogent Social Sciences* 12, no. 1 (2026): 2719286, 12–13, https://doi.org/10.1080/23311886.2026.2719286.

misunderstandings or miscalculations.[22] West and Azcárate Ortega give a reason not to treat that mechanism as automatic: norms still need acceptance and implementation, and weak trust or shared values can turn the process into competition for normative influence.[23]

Wiener's "meaning-in-use" is concerned with what happens after actors share the same vocabulary. The meaning does not sit only in the wording of a norm; practice gives it content.[24] In this case, "participation" could be spoken publicly without settling what conduct counted as participation. Fearon is dealing with a different problem. Public communication can still leave private information, capability and resolve undisclosed, while actors retain incentives to misrepresent them.[25]

Legal language also travels outside the legal test. NATO StratCom uses "lawfare" for hostile use of legal claims to damage or delegitimise an opponent, occupy its time or secure a public-relations advantage.[26] The label is not used here to infer Russian intent. JCOIE nevertheless gives a reason to take the public characterisation seriously, since actions and communications acquire informational aspects through the meaning observers assign to their features in context.[27] Russia, the United States and SpaceX did not necessarily use the same boundary when they spoke about commercial military support.

The Russia–Ukraine Case

Viasat found no evidence that the satellite itself or its supporting satellite ground infrastructure had been directly impaired. The destructive phase instead ran through a VPN misconfiguration into a trusted management segment. Legitimate management commands then left modems unable to reach the network. The affected partition was consumer-oriented. Several thousand customers in Ukraine lost service, along with tens of thousands elsewhere in Europe; directly managed government and mobility users did not. Its wording does not mean that no military user was affected.

---

22 Kuplic and Sawmiller, "Humanity on the Final Frontier," 235–37.
23 West and Azcárate Ortega, *Norms for Outer Space*, 26–27.
24 Antje Wiener, "Contested Compliance: Interventions on the Normative Structure of World Politics," *European Journal of International Relations* 10, no. 2 (2004): 189–92, https://doi.org/10.1177/1354066104042934.
25 James D. Fearon, "Rationalist Explanations for War," *International Organization* 49, no. 3 (1995): 395–97, https://doi.org/10.1017/S0020818300033324.
26 Sean Aday et al., *Hybrid Threats: A Strategic Communications Perspective* (Riga: NATO Strategic Communications Centre of Excellence, 2019), 38, 42.
27 Joint Chiefs of Staff, JCOIE, 4.

Ukrainian military users could still have been relying on the affected consumer-oriented partition.[28] MITRE later associated AcidRain with the outage and records destructive techniques against network devices.[29]

Viasat did not say what the operation was supposed to achieve. Governments did. The United Kingdom attributed the operation to Russia and said the Ukrainian military was believed to have been the primary target. [30] The United States described a Russian attack on commercial satellite communications networks intended to disrupt Ukrainian command and control. [31] The European Union tied the operation to the opening of the invasion, said it facilitated the military aggression and referred to "indiscriminate communication outages" across public authorities, businesses and users.[32] "Indiscriminate" is the language of that declaration. It is not used here as a finding under the law of attack.

Zhora did not describe the effect in the same way throughout 2022. His early account was of a "really huge loss in communications." By September he was talking more specifically about the satellite component, stressing that land lines remained the armed forces' priority and that coordination between forces had continued. Reports of radio jamming at the front made the difference between those accounts harder to read. Zetter added another complication. An unnamed source with recent Ukrainian military experience told her that Viasat was not widely distributed enough among Ukrainian forces for the outage to have caused major disruption.[33] The public material therefore establishes disruption more clearly than it establishes its military scale.

By October 2022, the Starlink record was still thin. On 14 October, Pentagon deputy press secretary Sabrina Singh confirmed that the Department was in communication with SpaceX and, when asked whether Starlink had mainly supported

---

28 Viasat, Inc., "KA-SAT Network Cyber Attack Overview," 30 March 2022, 1–2, https://www.viasat.com/perspectives/corporate/2022/ka-sat-network-cyber-attack-overview/.
29 MITRE ATT&CK, "AcidRain," Software S1125, version 1.1, last modified 15 April 2025, https://attack.mitre.org/software/S1125/.
30 UK National Cyber Security Centre, "Russia behind Cyber Attack."
31 Blinken, "Attribution of Russia's Malicious Cyber Activity."
32 Council of the European Union, "Russian Cyber Operations against Ukraine: Declaration by the High Representative on behalf of the European Union," press release 432/22, 10 May 2022, https://www.consilium.europa.eu/en/press/press-releases/2022/05/10/russian-cyber-operations-against-ukraine-declaration-by-the-high-representative-on-behalf-of-the-european-union/.
33 Kim Zetter, "Viasat Hack 'Did Not' Have Huge Impact on Ukrainian Military Communications, Official Says," *Zero Day*, 26 September 2022, including update at 3:30 p.m. PST, https://www.zetter-zeroday.com/viasat-hack-did-not-have-huge-impact/.

civilian communications or military command and control, replied: “Well, if I remember, I think it has been used for both.” The wording itself was cautious. Singh said “if I remember” and “I think” before describing Starlink as serving both civilian communications and military command and control. The answer put dual use into the public record without offering a breakdown of which functions, users or locations were military.[34]

Twelve days later Konstantin Vorontsov told the UN First Committee that US and allied military use of civilian and commercial space infrastructure amounted to “indirect participation in military conflicts.” He then warned that “Quasi-civilian infrastructure may become a legitimate target for retaliation.” Vorontsov delivered the warning during the First Committee’s thematic discussion on outer space. It was not a private exchange with SpaceX or a bilateral message to Washington. The statement was made before other state delegations and was also published by the Russian Mission.[35]

Karine Jean-Pierre was asked about that warning the next day. She answered in deterrent terms, saying an attack on US infrastructure would meet a response “in a time and manner of our choosing” and that the United States would pursue means to deter and hold Russia accountable for such attacks. [36] She did not answer Vorontsov on the law or accept his characterisation of commercial military support. The warning had been received; the response concerned what the United States might do if its infrastructure were attacked.

SpaceX’s public position emerged more slowly. Starshield appeared in December 2022 as a national-security business for government customers, extending SpaceX capabilities into higher-security communications and Earth observation.[37] In February 2023, Shotwell said Starlink had never been intended to be “weaponized” and described offensive uses by Ukraine as outside the agreement. Her distinction

34 U.S. Department of Defense, “Sabrina Singh, Deputy Pentagon Press Secretary, Holds a Press Briefing,” transcript, 14 October 2022, https://www.defense.gov/News/Transcripts/Transcript/Article/3189495/sabrina-singh-deputy-pentagon-press-secretary-holds-a-press-briefing/.
35 Vorontsov, “Statement … on Outer Space,” 26 October 2022.
36 The White House, “Press Gaggle by Press Secretary Karine Jean-Pierre En Route Syracuse, NY,” 27 October 2022, https://bidenwhitehouse.archives.gov/briefing-room/press-briefings/2022/10/27/press-gaggle-by-press-secretary-karine-jean-pierre-en-route-syracuse-ny/.
37 Courtney Albon, “SpaceX Forms ‘Starshield’ Business Unit to Focus on National Security,” *Defense News*, 5 December 2022, https://www.defensenews.com/battlefield-tech/space/2022/12/05/spacex-forms-starshield-business-unit-to-focus-on-national-security/.

was specific: “We know the military is using them for comms and that’s OK. But our intent was never to have them use it for offensive purposes.” She also said SpaceX had measures it could use, and had used, to limit that employment without identifying them. Shotwell did not identify those measures or explain how SpaceX distinguished an acceptable military communication from an offensive use in practice.[38] By June, a US defence official confirmed a Pentagon contract for Starlink satellite-communications capabilities in Ukraine.[39] The relationship was changing in public view, although the sequence does not tell us what caused each change.

Sevastopol is useful for a different reason because the disclosure arrived long after the event. Reporting around Walter Isaacson’s biography placed a Ukrainian request involving Sevastopol in 2022, but the episode only became public in September 2023. [40] Musk’s posts described the request as one to activate Starlink “all the way to Sevastopol.” He also said the relevant regions “were not activated.”[41] Those posts appeared in September 2023 after Isaacson’s account became public. They were not part of the public record in 2022.

Testing the Information Contest

Rid and Buchanan treat public communication as part of attribution itself, not as a separate step after assessment.[42] The EU declaration named Russia and connected the KA-SAT disruption to the opening of the invasion. It described “indiscriminate communication outages” affecting public authorities, businesses and users. The declaration also said the operation facilitated Russia’s military aggression and referred to expectations of responsible state behaviour.[43] The outage was already more than two months old when the declaration appeared. By May there were

38 Foust, “Shotwell: Ukraine ‘Weaponized’ Starlink.”
39 Jon Harper, “Pentagon Contracting with SpaceX’s Starlink to Provide Satellite Communication Capabilities for Ukraine,” *DefenseScoop*, 1 June 2023, https://defensescoop.com/2023/06/01/pentagon-contracting-with-spacexs-starlink-to-provide-satellite-communication-capabilities-for-ukraine/.
40 CBS News/AFP, “Elon Musk Says He Denied Ukraine Satellite Request to Avoid Complicity in ‘Major Act of War’ vs. Russia,” *CBS News*, 8 September 2023, https://www.cbsnews.com/news/elon-musk-ukraine-russia-war-starlink-satellite-denied-major-act-of-war/.
41 Elon Musk (@elonmusk), “There was an emergency request from government authorities to activate Starlink all the way to Sevastopol,” X, 7 September 2023, https://x.com/elonmusk/status/1699917639043404146; Elon Musk (@elonmusk), “The Starlink regions in question were not activated. SpaceX did not deactivate anything,” X, 7 September 2023, https://x.com/elonmusk/status/1699913329261813809.
42 Rid and Buchanan, “Attributing Cyber Attacks,” 7.
43 Council of the European Union, “Russian Cyber Operations against Ukraine.”

therefore two public accounts of KA-SAT: the technical account of what had happened to the service and the political account of who was responsible and why.

Shotwell was talking about permission. SpaceX would allow military communications and reject "offensive purposes", and she said the company had measures available to enforce that distinction.[44] Vorontsov was making a different kind of claim. He treated military use of commercial infrastructure as "indirect participation" and linked that characterisation to the possibility of retaliation.[45] The two positions did not meet at a common threshold.

Article 52 cuts across both. It asks about an object rather than a provider's contract or a state's political label. The object must make an effective contribution to military action, and attacking it must offer a definite military advantage in the circumstances at the time.[46] The Woomera Manual makes the same separation when it discusses military communications. A communication function may make an effective contribution to military action, but that does not by itself establish the definite military advantage expected from attacking the object. That second requirement still has to be assessed in the circumstances at the time, and the other rules governing attack continue to apply.[47] SpaceX could police its own service. That did not make its private boundary the legal one. More detail about a terminal, gateway or service setting could narrow the factual inquiry. Wiener's meaning-in-use places normative meaning in practice rather than in wording alone. [48]

Published guidance gives other actors more to work with than inference alone. Kuplic and Sawmiller connect guidance, state practice and dialogue with the development of common understandings and fewer misunderstandings or miscalculations.[49] The OEWG considered dual-use systems, commercial participation, megaconstellations used during military operations, humanitarian consequences from disabling space assets and cyber counterspace threats. No

44 Foust, "Shotwell: Ukraine 'Weaponized' Starlink."
45 Vorontsov, "Statement … on Outer Space," 26 October 2022.
46 Additional Protocol I, art. 52(2)–(3).
47 Beard and Stephens, *Woomera Manual*, 320–21.
48 Wiener, "Contested Compliance," 189–92.
49 Kuplic and Sawmiller, "Humanity on the Final Frontier," 235–37.

consensus substantive report followed. UNIDIR nevertheless identified several areas where states had moved closer together. It also found cross-regional support for using political commitments alongside legally binding measures.[50] The 2024 PAROS Group of Governmental Experts later adopted its report by consensus. [51] Fearon's problem survives all of this. Public discussion can expose a position without exposing the information behind it. A state may still conceal capability or resolve, and may still have reasons to present either misleadingly.[52]

Vorontsov spoke on 26 October 2022. The White House responded the following day, although its answer concerned the threatened attack on US infrastructure rather than Russia's definition of participation.[53] Starshield appeared in December. Shotwell discussed SpaceX's restrictions in February 2023, and the Pentagon contract became public in June. Put in a line, those events can look like stages in the same development. The public record does not establish that relationship. Battlefield requirements were changing during the same period, contracts were being negotiated and concerns about escalation were also present. The dates tell us what came first. They do not show that one event caused the next.

Paul's warning about measures of effectiveness matters for that reason. An observed response is weak evidence that an earlier message caused it when other explanations remain available.[54] The same problem applies when later evidence is read back into an earlier period. A later acknowledgement of military use may tell us more about a particular function or relationship, but it cannot alter what was publicly known in October 2022. Later disclosure improves the reconstruction of events; it does not move the disclosure itself backwards in time. Nor can it be used to assume that every satellite, gateway or terminal satisfied Article 52 at the earlier point in time.[55]

---

50 Almudena Azcárate Ortega and Sarah Erickson, *OEWG on Reducing Space Threats: Recap Report* (Geneva: United Nations Institute for Disarmament Research, 2024), 22, 30–33, 36, https://doi.org/10.37559/WMD/24/Space/01.
51 United Nations General Assembly, *Report of the Group of Governmental Experts on Further Practical Measures for the Prevention of an Arms Race in Outer Space*, A/79/364, 20 September 2024, 2.
52 Fearon, "Rationalist Explanations for War," 395–97.
53 White House, "Press Gaggle by Press Secretary Karine Jean-Pierre," 27 October 2022.
54 Paul, *Strategic Communication*, 119.
55 Additional Protocol I, art. 52(2)–(3).

Implications and Responses

SpaceX could still alter coverage and use conditions after Starlink had become operationally important in Ukraine.[56] Once a service reaches that point, a change in coverage is no longer an ordinary provider decision for the government relying on it. The contract can address that earlier. Notice before changes to coverage or functionality can be agreed before deployment. Permitted military uses can also be specified there. Some decisions may remain with the provider; others may require government approval or consultation between the two. Disclosure and escalation-sensitive decisions can be dealt with before they arise during an operation. Technical restrictions on coverage or functionality could also be agreed beforehand; geofencing is one option that could be considered. More state authority leaves another issue. A provider tied closely to military requirements may also become more closely embedded in military activity.

KA-SAT points to a different kind of failure. The vulnerable point sat in the path into the service rather than in destruction of the spacecraft itself. Defence users need alternative communications paths, and some separation between military functions and heavily civilian-dependent services may be sensible where feasible. The ICRC treats segregation of military and civilian use as one possible precaution rather than an absolute rule.[57] Viasat's account places the vulnerable point in a management function and service partition rather than the satellite as a whole.[58] Bi and Zhu's smallest-feasible-function approach keeps attention on dependencies and cascading effects at the functional level. [59] Complete separation is not always technically or economically sensible. Too much segmentation can also sacrifice the scale and redundancy that make commercial services useful.

56 Foust, "Shotwell: Ukraine 'Weaponized' Starlink."
57 International Committee of the Red Cross, *Constraints under International Law on Military Operations in, or in Relation to, Outer Space during Armed Conflicts*, working paper submitted to the Open-ended Working Group on Reducing Space Threats through Norms, Rules and Principles of Responsible Behaviours, UN Doc. A/AC.294/2022/WP.4, 3 May 2022, 5–6, https://www.icrc.org/en/document/constraints-under-international-law-military-space-operations.
58 Viasat, "KA-SAT Network Cyber Attack Overview," 1–2.
59 Bi and Zhu, "From Objects to Interdependence," 12–13.

Commercial operators also hold much of the technical knowledge needed to judge whether proposed rules are workable. UNIDIR notes that commercial entities make up around 80 per cent of the space economy, and the OEWG was open to industry and civil-society participation.[60] Their role does not make them public authorities, but service dependencies, technical feasibility and likely effects can be missed if they are absent from the discussion. West and Azcárate Ortega identify acceptance, implementation and trust as conditions on how norms operate in practice.[61]

Conclusion

KA-SAT makes "commercial SATCOM" look like too large a category. The satellite was not the point through which the destructive activity entered the service. Viasat instead traced it through a management function serving a consumer-oriented partition. Civilian disruption was clear. The extent of military disruption was not. That difference matters because Article 52 is concerned with the object in the circumstances at the time, not with whether a network can generally be described as dual-use or commercially owned.

Starlink exposed another kind of division. Ukraine could depend on the service without controlling it. Coverage still depended on SpaceX, as did the question of what counted as an unacceptable use. Those decisions mattered more once Ukrainian forces were already relying on the network. Russia then described Western military reliance on commercial infrastructure in much broader terms, as "indirect participation", and raised retaliation. The White House answered the threat to US infrastructure rather than accepting or rejecting that category. Shotwell's own line was narrower: military communications were allowed, offensive use was not. Nothing in the public exchange made that provider rule equivalent to Russia's category of participation.

60 Azcárate Ortega and Erickson, *OEWG on Reducing Space Threats*, 11.
61 West and Azcárate Ortega, *Norms for Outer Space*, 26–27.

Wiener and Fearon leave two different uncertainties in place. Wiener helps explain how participation can acquire content through repeated use and contest without settling into one agreed meaning. Fearon points to what public statements do not disclose: capability, resolve, thresholds and the reasoning behind them may remain private.[62] More was visible, but the actors still did not necessarily share the same information or the same interpretation.

The practical consequences begin before any of those legal questions are tested. A government relying on commercial SATCOM needs another communications path if coverage changes, the service fails or the provider refuses a use. It also needs the rules on coverage, functionality and permitted use worked out before an operation makes them urgent. Providers matter here because they hold technical knowledge about the service and its dependencies that cannot be read from ownership of the satellite alone.

By the end of the period, more of the dispute was on the record. Russia had stated how it understood military reliance on commercial infrastructure. The United States had stated how it would respond to a threat against its infrastructure, and SpaceX had explained some of the limits it placed on Starlink. That was useful, but it was not convergence. The remaining question was which of those different understandings, if any, would carry weight in future conduct.

[62] Wiener, “Contested Compliance,” 189–92; Fearon, “Rationalist Explanations for War,” 395–97.

Bibliography

Aday, Sean, Māris Andžāns, Una Aleksandra Bērziņa-Čerenkova, Francesca Granelli, John-Paul Gravelines, Mils Hills, Miranda Holmstrom, et al. *Hybrid Threats: A Strategic Communications Perspective*. Riga: NATO Strategic Communications Centre of Excellence, 2019.

Albon, Courtney. "SpaceX Forms 'Starshield' Business Unit to Focus on National Security." *Defense News*, 5 December 2022. https://www.defensenews.com/battlefield-tech/space/2022/12/05/spacex-forms-starshield-business-unit-to-focus-on-national-security/.

Azcárate Ortega, Almudena, and Sarah Erickson. *OEWG on Reducing Space Threats: Recap Report*. Geneva: United Nations Institute for Disarmament Research, 2024. https://doi.org/10.37559/WMD/24/Space/01.

Beard, Jack, and Dale Stephens, eds., with David Koplow. *The Woomera Manual on the International Law of Military Space Activities and Operations*. Oxford: Oxford University Press, 2024. https://doi.org/10.1093/law/9780192870667.001.0001.

Berrang, Svenja. "Does the Dual-Use of Space Objects Necessitate a New Geneva Convention?" *Case Western Reserve Journal of International Law* 57 (2025): 315–342. https://scholarlycommons.law.case.edu/jil/vol57/iss1/22/.

Bi, Chen, and Yiyang Zhu. "From Objects to Interdependence: How the Law of Armed Conflict Regulates Attacks on Dual-Use Information Infrastructure in Cyber and Information Warfare." *Cogent Social Sciences* 12, no. 1 (2026): 2719286. https://doi.org/10.1080/23311886.2026.2719286.

Blinken, Antony J. "Attribution of Russia's Malicious Cyber Activity Against Ukraine." U.S. Department of State, 10 May 2022. https://2021-2025.state.gov/attribution-of-russias-malicious-cyber-activity-against-ukraine/.

CBS News/AFP. "Elon Musk Says He Denied Ukraine Satellite Request to Avoid Complicity in 'Major Act of War' vs. Russia." *CBS News*, 8 September 2023. https://www.cbsnews.com/news/elon-musk-ukraine-russia-war-starlink-satellite-denied-major-act-of-war/.

Congressional Research Service. *Defense Primer: Operations in the Information Environment*. IF10771. Updated 29 November 2024. https://www.everycrsreport.com/reports/IF10771.html.

Council of the European Union. "Russian Cyber Operations against Ukraine: Declaration by the High Representative on behalf of the European Union." Press release 432/22, 10 May 2022. https://www.consilium.europa.eu/en/press/press-releases/2022/05/10/russian-cyber-operations-against-ukraine-declaration-by-the-high-representative-on-behalf-of-the-european-union/.

Deppe, Christoph, and Gary S. Schaal. "Cognitive Warfare: A Conceptual Analysis of the NATO ACT Cognitive Warfare Exploratory Concept." *Frontiers in Big Data* 7 (2024): 1452129. https://doi.org/10.3389/fdata.2024.1452129.

Fearon, James D. "Rationalist Explanations for War." *International Organization* 49, no. 3 (1995): 379–414. https://doi.org/10.1017/S0020818300033324.

Foust, Jeff. "Shotwell: Ukraine 'Weaponized' Starlink in War against Russia." *SpaceNews*, 8 February 2023. https://spacenews.com/shotwell-ukraine-weaponized-starlink-in-war-against-russia/.

Harper, Jon. “Pentagon Contracting with SpaceX’s Starlink to Provide Satellite Communication Capabilities for Ukraine.” *DefenseScoop*, 1 June 2023. https://defensescoop.com/2023/06/01/pentagon-contracting-with-spacexs-starlink-to-provide-satellite-communication-capabilities-for-ukraine/.

Hathaway, Oona A., Azmat Khan, and Mara R. Revkin. “The Dangerous Rise of ‘Dual-Use’ Objects in War.” *Yale Law Journal* 134, no. 8 (2025): 2645–2750. https://yalelawjournal.org/article/the-dangerous-rise-of-dual-use-objects-in-war.

International Committee of the Red Cross. *Constraints under International Law on Military Operations in, or in Relation to, Outer Space during Armed Conflicts*. Working paper submitted to the Open-ended Working Group on Reducing Space Threats through Norms, Rules and Principles of Responsible Behaviours, UN Doc. A/AC.294/2022/WP.4, 3 May 2022. https://www.icrc.org/en/document/constraints-under-international-law-military-space-operations.

Joint Chiefs of Staff. *Joint Concept for Operating in the Information Environment (JCOIE)*. Washington, DC: Joint Chiefs of Staff, 25 July 2018.

———. *Joint Publication 3-13: Information Operations*. Washington, DC: Joint Chiefs of Staff, 27 November 2012. Incorporating Change 1, 20 November 2014.

Kuplic, G. Blair, and Jonathan Sawmiller. “Humanity on the Final Frontier: Challenges in Applying International Humanitarian Law to Modern Military Space Operations.” *International Review of the Red Cross* 107, no. 928 (2025): 200–237. https://doi.org/10.1017/S1816383124000559.

MITRE ATT&CK. “AcidRain.” Software S1125, version 1.1. Last modified 15 April 2025. https://attack.mitre.org/software/S1125/.

Paul, Christopher. *Strategic Communication: Origins, Concepts, and Current Debates*. Santa Barbara, CA: Praeger, 2011.

Protocol Additional to the Geneva Conventions of 12 August 1949, and Relating to the Protection of Victims of International Armed Conflicts (Protocol I). 8 June 1977. 1125 U.N.T.S. 3.

Rid, Thomas, and Ben Buchanan. “Attributing Cyber Attacks.” *Journal of Strategic Studies* 38, nos. 1–2 (2015): 4–37. https://doi.org/10.1080/01402390.2014.977382.

UK National Cyber Security Centre. “Russia behind Cyber Attack with Europe-Wide Impact an Hour before Ukraine Invasion.” 10 May 2022. https://www.ncsc.gov.uk/news/russia-behind-cyber-attack-with-europe-wide-impact-hour-before-ukraine-invasion.

United Nations General Assembly. *Report of the Group of Governmental Experts on Further Practical Measures for the Prevention of an Arms Race in Outer Space*. A/79/364. 20 September 2024.

U.S. Department of Defense. “Sabrina Singh, Deputy Pentagon Press Secretary, Holds a Press Briefing.” Transcript, 14 October 2022. https://www.defense.gov/News/Transcripts/Transcript/Article/3189495/sabrina-singh-deputy-pentagon-press-secretary-holds-a-press-briefing/.

———. *Strategy for Operations in the Information Environment*. Washington, DC: Department of Defense, 2023.

Viasat, Inc. “KA-SAT Network Cyber Attack Overview.” 30 March 2022. https://www.viasat.com/perspectives/corporate/2022/ka-sat-network-cyber-attack-overview/.

Vorontsov, Konstantin. “Statement by Deputy Head of the Russian Delegation Mr. Konstantin Vorontsov at the Thematic Discussion on Outer Space (Disarmament Aspects) in the First Committee of the 77th Session of the UNGA.” Permanent Mission of the Russian Federation to the United Nations, 26 October 2022. Unofficial translation. https://russiaun.ru/en/news/261022_v.

West, Jessica, and Almudena Azcárate Ortega. *Norms for Outer Space: A Small Step or a Giant Leap for Policymaking?* Space Dossier 7. Geneva: United Nations Institute for Disarmament Research, 2022. https://doi.org/10.37559/WMD/22/Space/01.

The White House. “Press Gaggle by Press Secretary Karine Jean-Pierre En Route Syracuse, NY.” 27 October 2022. https://bidenwhitehouse.archives.gov/briefing-room/press-briefings/2022/10/27/press-gaggle-by-press-secretary-karine-jean-pierre-en-route-syracuse-ny/.

Wiener, Antje. “Contested Compliance: Interventions on the Normative Structure of World Politics.” *European Journal of International Relations* 10, no. 2 (2004): 189–234. https://doi.org/10.1177/1354066104042934.

Wither, James K. “Making Sense of Hybrid Warfare.” *Connections: The Quarterly Journal* 15, no. 2 (2016): 73–87. https://doi.org/10.11610/Connections.15.2.06.

Zetter, Kim. “Viasat Hack ‘Did Not’ Have Huge Impact on Ukrainian Military Communications, Official Says.” *Zero Day*, 26 September 2022. https://www.zetter-zeroday.com/viasat-hack-did-not-have-huge-impact/.